

Operator Packages, Proposer Strength, and Construction-Family Plateaus in Office-Scale Verified Search

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The changes in version 4 are listed after the abstract.

Abstract

Verified search, in which a language model proposes programs, a hard evaluator scores them, and selection keeps the best, has recently moved mathematical records; controlled ablations of the proposer-side components remain rare. We instrument a minimal FunSearch-style loop at office scale (a 30B local model on a laptop, 120–600 verified samples per run) with three operator packages: a schematic notebook the model writes and carries instead of verbatim elites, a named obstacle, and behavioural repulsion from constructions already found. On nine construction problems from a public repository, the complete 2^3 factorial with two replicates favours the primary contrast in a nominal two-stage analysis: the composition closes more of the seed-to-record gap (+0.196 [+0.059, +0.343]; nominal pooled $p = 0.023$, stage-combination $p \approx 0.08$; median per-problem effect +0.045; robust to removing the seed floor from the metric). Repulsion raises construction-hash diversity in eight of nine problems (+0.31, one-sided $p = 0.0039$; partly a manipulation check, since the operator optimizes a relative of this quantity). The factorial finds no positive memory-by-repulsion interaction and bounds any such interaction to about ± 0.04 : the gain decomposes additively, the notebook package carrying the largest descriptive share, and memory+repulsion is the only arm that never collapses (0 of 18 runs), with a mean 0.025 below the full composition’s, a difference unresolved at $n = 9$. A frontier proposer under the identical loop reaches in tens of samples what the local model does not in hundreds, and in single scoping runs its gains arrive without the operators (differences at most 0.001 at equal grid). The searches stall after closing 85–92% of the gap toward the best published construction on the flagship problem, and a later analysis (Section 5, added in v3) reads that plateau as a resolution limit: the reference value is a continuum limit and the unaided long-run finals already carry its family’s structure. In the registered family-hint test the family named in words is adopted and loses; handed as code, it is optimized, but our best finite-grid encoding stays below the plateau reached unaided. We release the harness, every candidate, and the dated pre-registrations.

Changes in version 4. Corrections; no registered test or verdict changes. (1) Repulsion raised construction-hash diversity in eight of nine problems (Heilbronn -0.04) and in 16 of 18 replicates, not “everywhere” or “in every problem and replicate” as versions 1–3 said; $p = 0.0039$ is one-sided and is the p of one adverse cell, not the minimum attainable ($1/512$). (2) On beat-the-average the three values compared lie at 85–92% of the seed-to-reference gap, not 92–98%. (3) The abstract, Sections 6 and 7 and the Conclusion now agree with the note added in version 3 (the plateau read as a resolution limit): the family signature is absent in the three registered no-hint runs, but the same classifier labels five of the sixteen unaided factorial finals in-family. (4) Grids are named in the proposer comparison: Claude’s

0.3897 is at $L=20000$; at $L=5000$ its composed arm reached 0.3872 in a later run, which the 27B matches; the frontier-strength operator differences are at most 0.001 at equal grid. (5) The 27B was served with vLLM on a rented H100, not locally. (6) The note is labelled as added in version 3, not version 2; “the limit was not resolution” is replaced; three proposers, not two; nine problems, not six; 68 pages (67 problems); one typo; the provenance table gains the 27B battery and reports the interaction as not supported; references updated. (7) Data availability now points to the Zenodo deposits (Appendix A), and a statement on the use of an AI assistant was added.

1 Introduction

FunSearch [14] and AlphaEvolve [11] demonstrated that a language model inside an evolutionary loop with a hard evaluator can produce genuinely new mathematical constructions. The systems that followed improved the *scaffolding*: sample-efficient archives and novelty rejection [7], verbal reflections as search gradients [18], stronger databases and routing. Two questions have stayed unmeasured. First, *which parts of what the proposer is asked to think* (its memory of the lineage, its statement of the obstacle, its relation to what was already found) change the search, and by how much? The prompts of the published systems show elite programs verbatim and vary everything else. Second, *how much of the outcome is the proposer itself*, holding the loop fixed? Published systems change model and scaffold together.

This paper measures both, at a scale any laboratory can afford, with the discipline our earlier studies [12] found necessary: every battery pre-registered before running, dated amendments, exact paired tests, and negative results reported at the same volume as positive ones. The operators we test are not arbitrary: they are the three interventions that survived a two-paper program on open-ended generation: a *schematic memory* (the model’s own compressed recap outperformed verbatim context on document-level integration), a *standing question*, and *repulsion* from the already-produced, which in weight space collapsed without an anchor and in behaviour space is the natural tail-preserving form.

Contributions. (1) A pre-registered program of cognitive operators inside a fixed verified-search loop, grown from a $5 \times 6 \times 2$ ablation to the complete $2^3 \times 9 \times 2$ factorial: the composition reaches +0.196 (nominal $p = 0.023$ over the two-stage sequence; robust to the metric’s seed floor), behavioural repulsion buys construction-hash diversity in eight of nine problems and 16 of 18 replicates ($p = 0.0039$), and no single package avoids every failure mode. (2) A revision, by the completed factorial, of this program’s recurring *anchoring pattern*: the factorial finds no positive memory $\times$ repulsion interaction and bounds any such interaction to about ± 0.04 of frac; what remains of anchoring is descriptive *collapse prevention*, not superadditive gain. Repulsion without memory still breaks (three collapses in eighteen runs; four when paired with agenda), while memory+repulsion never does, echoing unanchored preference optimization in weight space. (3) A *scaling inversion*: from a 120-sample tie, the composed arm climbs to 0.3816 at 600 samples while repulsion alone stalls at 0.3514: diversity without accumulated direction does not compound. (4) A controlled *proposer comparison*: local 30B versus a frontier model under the identical loop and budget, on the same problem, locating the residual gap in the proposer rather than in the loop, with a frontier-strength scoping run in which the single-run operator differences are at most 0.001 at equal grid. (5) An honest calibration of office-scale verified search: the composed arm closes a median 0.71 of the seed-to-reference gap (per-problem 0.10–0.99; two problems sit at plateaus every arm shares), the tested grid and verifier-time changes did not improve the result, a stronger proposer raised and accelerated the flagship plateau without closing the final gap. A note added in version 3 (Section 5) reads the flagship plateau as a resolution limit of the verifier’s grid rather than a family boundary: the reference value is a continuum limit, finite-grid members of its family cap near 0.389, and the loop finals already carry the family’s structure.

2 Related work

Verified search systems. FunSearch evolved single functions with small code models and millions of samples [14]; AlphaEvolve evolves whole files with frontier ensembles and thousands [11], and its repository of 67 problems with official verifiers [4] is the substrate of our experiments. ShinkaEvolve reports record-level results with ~ 150 samples using novelty-based rejection and bandit model selection [7]; ReEvo uses model-written reflections as “verbal gradients” [18]. Our schematic notebook differs from ReEvo’s reflections in being a persistent, compressed lineage memory that *replaces* verbatim elites rather than commenting on pairs. A recent wave adds persistent memory to evolutionary code search: ε -MemEvo transfers compact strategy summaries across tasks with an adaptive injection gate and reports gate ablations [9]; EvoMem reuses mutation knowledge across runs [15]; MEMOIR shares algorithmic and failure-mode summaries across search branches [6]. What remains absent, and what we contribute, is a complete factorial, within one task suite and under a fixed local proposer, of a schematic-notebook package, an obstacle line, and a construction-hash duplicate control. LoongFlow [16] composes planning, hybrid memory and MAP-Elites niches into one system and reports large efficiency gains; it is the closest architecture to our composed arm, and exactly the kind of system whose components an ablation like ours aims to isolate.

Why diversity needs help. Without selection pressure, LLM mutation chains collapse into attractor regions [5]; RL-tuned models lose $\text{pass}@k$ coverage relative to their base [19]; and coverage under repeated sampling grows log-linearly only when a verifier exists [2]. Our behavioural-repulsion operator is novelty search [8, 10] transplanted into the proposer’s prompt, with the construction itself (not an embedding) as the behaviour descriptor.

Automated researchers at industrial scale. Concurrently with this work, Chen et al. [3] ran LLM “automated alignment researchers” that propose and train post-training fixes for ten alignment failures. Their harness independently converges on the organs studied here (a persistent memory file carried across fresh sessions and a shared finding forum, whose removal costs the most headroom in their ablation), and their telemetry shows each failure collapsing into a single dominant method family while idea diversity stays flat or declines as scores climb. A novelty-rejection variant raises the novelty of accepted methods but yields mixed downstream gains, mirroring the dissociation we certify here between diversity and value. Their setting is complementary to ours: mitigating a well-characterized failure is a task whose known-good region already exists in the literature, precisely the regime in which loops of this family excel; the construction-family plateau we document marks the boundary of that regime.

3 The harness, the operators, the problems

Loop. A minimal FunSearch-style loop: K islands, each holding its best programs; per generation, per island, a prompt is built and S candidates sampled; each candidate runs in a sandbox (time and memory limits) and is scored by the problem’s verifier; islands keep bounded elites and migrate their best every third generation. The proposer is **Qwen3-Coder-30B-A3B-Instruct** (8-bit, MLX, on a MacBook M5 Pro) prompted through its chat template; Section 5 swaps in Claude Opus 5 through an API under the identical loop. Every candidate is logged: code, score, error, generation, island, and a behaviour signature (a hash of the rounded construction it outputs).

Operators. **A** (baseline): the prompt shows the statement and the island’s top-3 programs with scores, FunSearch-style. **B** (schematic notebook): each generation the model writes a

frac (pooled over 2 replicates)	A	B	C	D	BC	BD	CD	E
circle packing 26	0.703	0.974	0.976	0.000	0.989	0.752	0.980	0.992
ring loading 15	0.444	0.767	0.808	0.808	0.731	0.808	0.808	0.808
beat-the-average	0.181	0.384	0.519	0.608	0.661	0.789	0.540	0.709
autocorrelation C_1	0.059	0.156	0.041	0.202	0.014	0.212	0.000	0.104
isosceles-free 64^2	0.560	0.560	0.560	0.399	0.560	0.560	0.560	0.560
sum-difference III	0.392	0.329	0.253	0.158	0.314	0.327	0.069	0.362
max-min distance 16	0.169	0.992	0.372	0.993	0.919	0.793	0.247	0.708
Heilbronn triangle 11	0.672	0.431	0.507	0.540	0.371	0.477	0.619	0.700
180! into 180 factors	0.774	0.774	0.774	0.774	0.774	0.774	0.774	0.774
mean frac	0.439	0.596	0.534	0.498	0.592	0.610	0.511	0.635
mean construction-hash diversity	0.41	0.53	0.52	0.72	0.59	0.68	0.80	0.70
collapses (frac < 0.05, of 18 runs)	2	1	1	3	2	0	4	1

Table 1: The complete 2^3 factorial over nine problems, pooled over two independent replicates (120 verified samples per run). frac = fraction of the seed-to-record gap closed. E has the best mean and is best or tied-best in five of nine problems; D alone is bimodal: best on wide valid manifolds (pmfs, sequences), catastrophic where validity is fragile (packing geometry, constrained integer sets); BD is the only arm that never collapses, CD the one that collapses most; two problems (isosceles-free, 180!) are family plateaus where every arm meets the same wall.

≤ 12 -line notebook (what worked, what failed and why, the open question) from the elites and the recent failures; the prompt then shows the notebook and only the single best program. **C** (agenda): one model-written line naming the structural obstacle of the current best, appended to the prompt. **D** (behavioural repulsion): candidates whose construction duplicates an island member’s are rejected, and the prompt lists already-found constructions as forbidden. **E**: B+C+D. We refer to these as operator *packages*: each bundles several changes at once (B replaces three verbatim elites with one, adds failure summaries and an open question, and spends an extra call; D combines a prompt-level forbidden list with a post-verification duplicate filter), so the factorial isolates packages, not single cognitive mechanisms; a finer within-package decomposition is registered as future work.

Problems. Nine construction problems ported from the official repository notebooks, chosen for cheap exact verification and known headroom. The original battery: circle packing (26 circles, maximize the sum of radii), ring loading ($m=15$, exact 2^m verification), beat-the-average (a pmf maximizing $P[X_1+X_2+X_3 < 2X_4]$), the first autocorrelation inequality (minimize an upper bound on C_1), isosceles-free subsets of the 64^2 grid, and sum-difference III. A pre-registered extension, run before any $n=9$ analysis, added three: the max-min pairwise-distance ratio for 16 points in the plane (minimize), the Heilbronn triangle problem for 11 points (maximize the smallest triple area), and splitting 180! into 180 factors maximizing the smallest (exact big-integer verification). Each run’s budget is 120 verified samples (10 generations $\times$ 2 islands $\times$ 6); the primary metric *frac* is the fraction of the seed-to-record gap closed, *auc* its mean over the budget (sample efficiency), *div* the share of distinct behaviours among valid candidates.

4 The factorial

Pre-registered contrasts (exact paired sign-flip over problems, cells = problems, pooled replicates). The original six-problem battery put the primary contrast at the design’s resolution floor (+0.199, $p = 0.0625$; with six cells, one adverse problem and one tie the exact test cannot go below $4/64$); the prospectively registered nine-problem extension supports it: **F1** (primary), E vs A on frac: +0.196 [+0.059, +0.343], $p = 0.023$ over the combined nine problems. We report

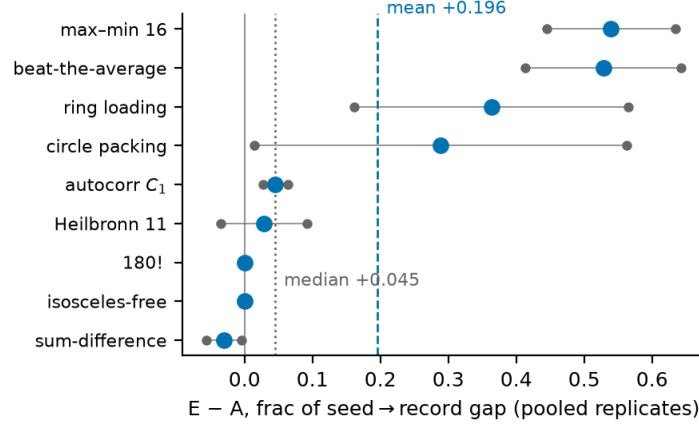


Figure 1: The primary contrast, problem by problem: $E - A$ on the fraction of the seed-to-record gap closed (pooled over two replicates). Small grey dots are the two replicates (lines join them); the large dot is their mean. The pooled mean is $+0.196$; the median, $+0.045$; leave-one-problem-out means stay between $+0.153$ and $+0.224$. Within-problem replicate spread is large on several problems (circle packing: $+0.01$ and $+0.56$), which the problem-level test does not see.

the stages separately, since the combination is a two-look sequence rather than an independent confirmation: the three extension problems alone are positive but underpowered ($+0.19$ on average, $p = 0.25$, carried mostly by max-min distance), and a Fisher combination of the stage p -values ($0.0625, 0.25$) gives $p \approx 0.08$, so the pooled 0.023 should be read as nominal accumulated evidence. The result is robust to the outcome definition: with the seed floor removed the contrast is unchanged ($+0.196$, $p = 0.023$), and excluding the two problems where every arm meets the same plateau it rises to $+0.252$ at the same p ($n = 7$). The effect is also heterogeneous (Figure 1): the median per-problem effect is $+0.045$, leave-one-problem-out means span $+0.153$ to $+0.224$, with large gains on four problems, little on four, and one small loss. **F4**, D vs A on construction-hash diversity: $+0.307$ [$+0.149, +0.477$], $p = 0.0039$ (one-sided), positive in eight of nine problems (Heilbronn -0.04) and in 16 of 18 replicates. **F6** (the anchor), E vs D: $+0.137$ [$-0.047, +0.382$], $p = 0.156$: not supported as a pairwise contrast; the factorial below resolves what it was measuring. **F2** (auc): $+0.115$, $p = 0.070$: the composed arm’s advantage builds late. (The operators’ measured overhead is small: one extra call per island-generation per operator, ~ 220 characters each, 1–3% of proposer output; Section 9 gives the accounting.)

The completed 2^3 . Three pairwise arms (memory+agenda, memory+repulsion, agenda+repulsion) over the same nine problems and two replicates complete the cube ($8 \text{ cells} \times 9 \times 2$; pre-registered before running). The effect-coded analysis (exact sign-flip over problems, 2^9) finds no evidence for the primary interaction: memory $\times$ repulsion $\beta = +0.003$ [$-0.039, +0.035$], $p = 0.47$ one-sided; agenda $\times$ repulsion is null (-0.007 , $p = 0.79$); and no main effect survives BH correction: schematic memory is the largest single term ($\beta = +0.057$ [$+0.014, +0.105$], raw $p = 0.047$, $q = 0.14$), agenda ($+0.016$) and repulsion ($+0.011$) near zero on frac. The decomposition is exact ($E - A = 2(\beta_M + \beta_A + \beta_R + \beta_{MAR}) = 0.196$): the nominal pooled composite gain splits as roughly 58% notebook, 16% agenda, 12% repulsion and 14% three-way (a descriptive decomposition with wide per-term intervals); at $n = 9$ the total is directionally supported, the parts remain unresolved. Two patterns in the cube are worth the trip. First, each operator has one job (Table 1, Figure 2): the four memory-on cells hold the four best mean fracs, while every repulsion-on cell sits at 0.68–0.80 construction-hash diversity against 0.41–0.59 without it; directionally, B-on cells hold the higher means and D-on cells the higher hash diversity, with the caveat that no main effect survives correction. Second, the anchor lives in the tail, not the mean: memory+repulsion is the only arm that never collapses

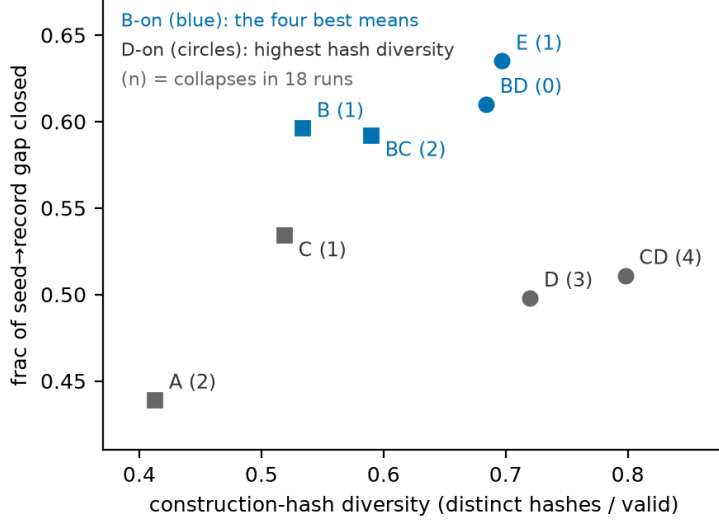


Figure 2: The completed 2^3 at a glance (cell means over nine problems, two replicates). Memory-on cells (blue) hold the four best gap fractions; D-on cells (circles) hold the highest construction-hash diversity; the two axes barely interact ($\beta_{M \times R} \approx 0$). Collapse counts in parentheses: memory+repulsion is the only arm that never collapses; agenda+repulsion collapses most.

(0 of 18 runs, against 3 for repulsion alone and 4 for agenda+repulsion, the cube’s worst), so what memory buys repulsion is insurance against starving the island, not superadditive mean gain. Agenda, decomposed at last, pays for neither: near-zero on frac alone, and the worst collapse count when paired with repulsion. Exploratory pairwise contrasts back the parsimony reading: E vs BD +0.025 ($p = 0.55$, median 0.000), E vs B +0.039 ($p = 0.47$), BD vs B +0.014 ($p = 0.81$), all null at $n = 9$. The collapse asymmetry is robust to the threshold: at $\text{frac} \leq 0$, < 0.01 , < 0.05 and < 0.10 the counts stay 3/3/3/3 for repulsion alone and 3/4/4/4 for agenda+repulsion against 0/0/0/0 for memory+repulsion, and no run anywhere produced zero valid candidates, so these are performance collapses, not validity collapses.

The heterogeneity is the finding the factorial adds to the literature: *where* the valid manifold is wide (a pmf, a sequence of pairs), repulsion alone is the best single operator; where validity is fragile (non-overlapping circles, exact integer constraints), it is associated with performance collapses (both replicates of circle packing), and the composed arm inherits most of the benefit with none of the catastrophes. No single package avoids every failure mode; memory+repulsion was the only configuration that avoided all of them in these runs.

5 Scale, proposers, and the wall

We then took the problem with guaranteed headroom, beat-the-average, where AlphaEvolve’s own reported value (0.3890) sits below the best known (0.400695), and ran a pre-registered ladder. (i) *Budget*: at 600 samples the composed arm reaches 0.3816 while repulsion alone, which had tied it at 120 samples, stalls at 0.3514: diversity without accumulated direction does not compound. (The same $5 \times$ extension on max–min distance 16 left both local arms *worse* than their own 120-sample runs on a fresh seed: seed variance dominates budget there too.) (ii) *Grid*: refining the pmf grid from $L=5000$ to $L=20000$ at equal budget makes the local proposer *worse* (0.3561): a finer grid did not help this proposer’s search (the note below explains why the ceiling itself rises only slowly with the resolution). (iii) *Proposer*: Claude Opus 5 under the identical loop reaches 0.3897 (at $L=20000$) within ~ 100 samples (just above the repository’s published value, obtained with 2025-era proposers; we read this as locating the wall, not as ranking systems

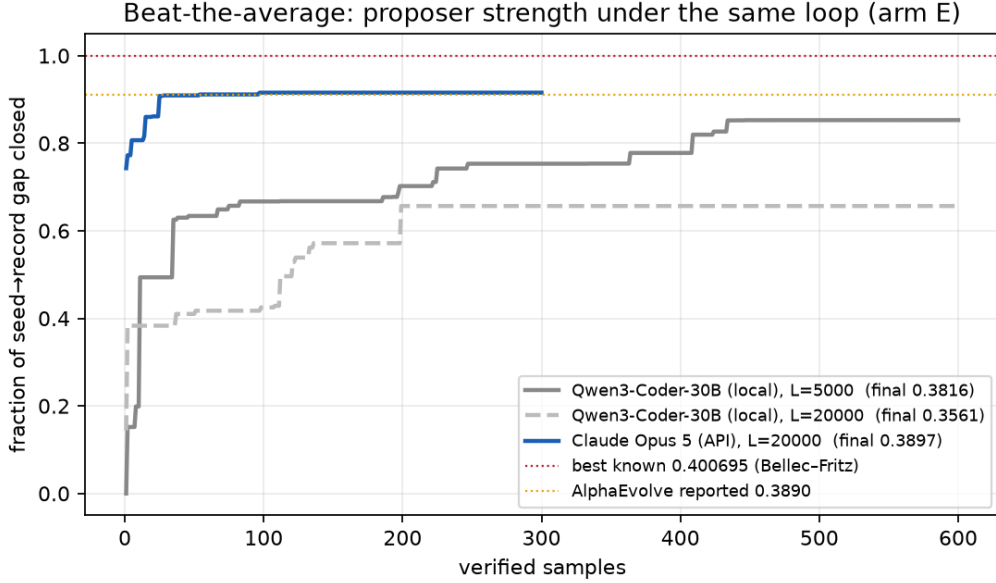


Figure 3: Beat-the-average under the same loop (arm E): fraction of the seed-to-record gap closed per verified sample. The local 30B at two grid resolutions; Claude Opus 5 at $L=20000$; dotted lines mark AlphaEvolve’s reported value (0.3890) and the best known (0.400695, Bellec-Fritz).

across model generations) and plateaus there for 17 generations. (iv) *Verifier time*: tripling the sandbox budget to 60 seconds, so heavier optimizers can run, moves the plateau by +0.0002. (v) *Proposer generation* (pre-registered on 1 September 2026, after version 1): a current-generation open-weight proposer, Qwen3.8-27B served with vLLM on a rented H100 (search and verifier local) in non-reasoning mode with a 3000-token budget (the earlier proposers had 1400), under the identical 600-sample loop at $L=5000$ reaches 0.3876, between the local model (0.3816) and Claude at $L=20000$ (0.3897); at $L=5000$ Claude’s composed arm reached 0.3872 in a later run, so at equal grid the 27B matches it. Its best construction is the same sparse-atom family, now with the exact objective, L-BFGS weight refinement and coordinate descent on atom positions; 60% of its valid samples exceed the local model’s whole plateau, its generation zero shows no near-plateau (best 0.351, like the local model and unlike Claude), so the gain is the search’s, and its validity rate is 61% (wrong-length outputs and timeouts). The wall survived a newer proposer by 0.013.

The constructions are legible. The local model’s best is a bimodal sparse measure of 16 atoms (half the mass on $\{0, 1, 2, 3\}$, geometrically spaced atoms at the top), with the strategy stated in comments. Claude’s best implements an exact sparse-atom objective with an analytic gradient and a multiplicative-weights refinement. The frontier proposer and AlphaEvolve’s published value sit within 0.0007 of each other; the local model, at $L=5000$, stalls about 0.008 behind them. All three lie at 85–92% of the seed-to-reference gap toward a reference whose construction we read, at the time, as lying outside the sparse-atom family every language model proposes (the note below revises that reading). More precisely: the candidates of all three proposers are small finitely supported measures found numerically, while the Bellec and Fritz [1] construction is a multiscale, recursively structured sequence of discrete measures whose analysis shows a finitely supported measure cannot be optimal; the family-hint conditions below name that structure. Within the settings tested, grid and time changes did not move the result and two stronger proposers moved it without closing the gap; these plateaus motivated the hypothesis that what remained was the construction family, an idea. The family-hint experiment below tests and qualifies that interpretation.

Note added in v3: the wall is the grid’s resolution. A later analysis of the archived beat-the-average finals (3–4 September 2026; appendix and scripts in Zenodo 10.5281/zenodo.22713807, Appendix A) changes the reading of this plateau. The reference value 0.400695 is the continuum limit of a self-similar scale mixture [1, Prop. 3.3]; the single dyadic ladder gives 0.3804 in the continuum and 0.3814 with optimised weights. Hand-written finite-grid members of the mixture family score 0.3850 at $L=5000$, 0.3867 at $L=20000$ and 0.3888 at $L=4,000,000$, rising slowly with the resolution. The loop finals of the 27B at $L=5000$ (0.3876) and of the frontier proposer at $L=20000$ (0.3899, a later run) already carry the family’s structure (a dyadic ladder at the top, a smaller copy of the ladder near zero, and split atoms that break the ties the continuum construction exploits) and exceed our hand-written members at the same grid. The structure is read by inspection of the atoms: under the signature classifier registered in Section 6, of these finals only the local 30B’s (0.3816, 600 samples) receives the ladder label, while the 27B’s and the frontier proposer’s fall in the sparse-other class; so “no unaided run matched the signature” (the three registered no-hint runs, by the classifier) and “the finals carry the structure” (the long-run finals, by inspection) are both true. The family was therefore not outside what the proposers produce; what none of them can reach on a finite grid is the continuum limit. This also explains the twist of Section 6: our strongest hand-written encoding, 0.374, sits below the mixture’s 0.385 at $L=5000$, so the family handed as code was a weaker member of a family the loop already had. On this problem the binding constraint is the resolution of the grid, not an idea; the other two shared plateaus (isosceles-free, 180!) remain candidate family boundaries. To our knowledge, no language-model system has reported more than about 0.390 on this problem. On other problems (circle packing, max–min distance) a bandit study of compute allocation over evolutionary trajectories with open models measured the same lock-in to the first family a run reaches [17].

A second problem sharpens where such ideas live. On max–min distance 16, the frontier proposer under the same loop emits *in generation zero*, before any search feedback, a minimax program (SLSQP on $\min t$ s.t. $d_{ij}^2 \leq t$, $d_{ij}^2 \geq 1$, analytic Jacobians) whose value lies marginally below the repository’s published one; every valid sample across the run converges to the same value. The loop verified this construction; it did not discover it. We treat the datum as scope, not as a record claim: the operators structure search and protect its tails, but they do not inject construction knowledge the proposer lacks. The pre-registered scoping runs make the same point from the other side, on all three registered problems. On beat-the-average, the bare, memory-only and repulsion-only arms land within 0.001 of one another (0.3871, 0.3866, 0.3876; single runs, estimation only) and within 0.001 of the composed arm at the same grid (0.3872; the 0.3897 above is at $L=20000$). On max–min distance and on the Heilbronn problem every arm converges to an identical value (12.889230 and 0.036530 respectively; the Heilbronn reference in our registry floors the reported figure, so that value reaches the frozen reference and supports no exceedance claim). At frontier strength the single-run operator differences are at most 0.001 at equal grid everywhere we measured them: the nominal gains are a local-proposer phenomenon, and whether the collapse insurance survives at this strength remains untested (single runs; the frontier proposer’s own run-to-run noise is unestimated).

6 The family-hint test: verbal specification versus executable seed

The registered family-hint experiment (five conditions $\times$ three seeds, 120 samples each, memory+repulsion arm, beat-the-average) asks what it takes to move the search into the Bellec and Fritz [1] family: (a) no hint; (b) a generic hint; (c) a structural hint naming the family without parameters; (d) a crude in-family seed program (initial score 0.2734); (e) a *strong in-family seed*, our best tested finite-grid instantiation of the family (0.3742; on this grid with strict in-

condition	seed 11	seed 12	seed 13	mean	finals in-family
no hint	0.3496	0.3607	0.3463	0.3522	0/3
generic hint	0.3512	0.3381	0.3086	0.3326	0/3
structural hint	0.3284	0.3383	0.3140	0.3269	3/3
crude in-family seed	0.3554	0.3679	0.3520	0.3584	2/3
strong in-family seed	0.3762	0.3739	0.3743	0.3748	3/3

Table 2: Family-hint test: final best per seed. In-family membership by the preregistered family-signature classifier (at most 50 support atoms, mass ≥ 0.3 on $\{0..3\}$, atoms in $\geq 3 \log_{10}$ distance-from-top bands; the structural hint instructs exactly this structure, so the classifier is partly a manipulation check of instruction-following). With three seeds per condition every contrast is descriptive (the smallest one-sided sign-flip p at $n=3$ is 0.125).

equality, notably *below* the 0.3816 plateau of the unaided 600-sample run, a fact measured and registered before launch; the condition supplies an author-implemented finite approximation, not the theoretical multiscale construction itself).

The answer came with a twist (Table 2). The structural hint *worked as an instruction and lost as a search*: its final constructions carry the family signature, so the model obeyed, entered a family that is inferior at this grid, and paid about 0.025 for it. Handed the family as code, the loop optimizes it (+0.085 from the crude seed, above the no-hint endpoint in three of three seeds) and stays loyal to it (no strong-seed run deserted to the sparse constructions of the unaided runs). No unaided candidate in the three registered no-hint runs matched the family signature. Two caveats bound the reading: the historical 0.3816 comes from a different battery under strong seed variance, so the strong seed’s 0.375 exceeds the *contemporary* no-hint 0.352 descriptively while trailing the historical best; and the seed conditions enter with different initial scores, so final values mix family advantage with starting-point advantage (matched-score sparse seeds and paraphrased or deliberately wrong hints are the registered next controls). Within those bounds the registered hypothesis was not supported: handing the idea, in words or in this finite encoding, did not cross the last mile under this budget; our tested finite-grid encoding of the idea was not sufficient to close the gap, which leaves open whether the theoretical family, a richer encoding, or a larger budget would. What the experiment does establish is a division of labour: under these runs the loop transported and optimized the idea it was handed and did not produce its signature unaided, and instructing a construction is not inducing a competitive implementation of it, the third appearance of that pattern in this program (the continuity connective of the companion study; the frontier scoping runs; now the structural hint).

7 Discussion

What office scale buys. With 120–600 verified samples, a local model and one laptop: a median 0.71 of the seed-to-reference gap closed by the composed arm across nine problems (per-problem 0.10–0.99), legible programs, and, with a frontier proposer at $\sim$ US\$50 of API, parity with the values the large systems report on this problem. The published record-movers used frontier ensembles with thousands of samples [11], millions with small models [14], or open-source loops with strong API models [7]; the band we measure (local proposer, hundreds of samples, controlled operators) was unoccupied, and it is the band most laboratories can afford.

The operator ledger. E had the highest mean and one collapse; memory+repulsion was the only zero-collapse arm, at a mean within 0.025 of E’s and without the agenda’s extra call, so the parsimonious operational recipe these data support is memory+repulsion, with the full composition as the highest-mean variant. The mechanism is visible in the notebooks: the notebook package carries lineage information across generations and the duplicate control

spends the budget on genuinely different constructions. Alone, each fails somewhere: memory and agenda are inert on problems the model already saturates; repulsion is associated with performance collapses on fragile manifolds. This is a recurring collapse-prevention pattern, consistent with the anchoring hypothesis, that this program has now observed across narrative generation, weight-space preference optimization, and prompt-space search.

The proposer comparison and the plateau. On this one problem, the frontier proposer is far more efficient in verifier evaluations (+0.008 absolute over the local model’s 600-sample best, at $L=20000$ against $L=5000$, and +0.006 at equal grid in a later run; reached with roughly $20\times$ fewer verified samples), and everything we tried after it moved little (grid, time, more samples: +0.0002). This is a sample-efficiency comparison on one problem under one harness, not yet a proposer-strength curve; replications on further problems were planned but not run, and reasoning effort, ensembles, alternative representations and thousand-sample budgets all remain untested explanations. The shared plateau suggests, consistently with our consolidation results [13], that search over a fixed prior rapidly collects what the prior’s construction families contain and then stalls at the best family member, a hypothesis to test rather than assert. Section 6 put that hypothesis to a registered test; the short answer is that a verbal specification of the reference family was followed and lost, while an executable in-family seed was optimized, and no unaided candidate in the three registered no-hint runs produced the family signature (applied later to all archived finals, the same classifier labels five of the sixteen unaided 120-sample factorial finals in-family). (The note in Section 5 revises this for the flagship problem: the long-run finals do carry the family’s structure, and the plateau is the grid’s resolution limit.) This is the question Georgiev et al. [4] report answering with an expert in the loop, from the other side.

Design rules. For verified search at small scale: pair memory with repulsion (the zero-collapse arm; the full composition adds +0.025 of mean at the cost of the agenda’s extra calls, a difference our data cannot distinguish from zero), and never run repulsion alone on fragile manifolds; keep the behaviour descriptor construction-derived (a rounded construction hash, not an embedding; no symmetry canonicalization); spend on the proposer before spending on samples; treat plateaus as candidate family boundaries and test them (grid, time, proposer and hints in pre-registered steps) rather than declaring them.

8 Experimental provenance

Every battery was registered before running; the combined analyses are sequential, and the table makes the order of looks explicit.

battery	registered	data already seen	primary	status
F ($5\times 6\times 2$)	2026-08-22	none	$E>A$ frac	$p = 0.0625$ (floor)
F-ext ladder	2026-08-23	F	budget/grid/proposer	descriptive
F-ext2 ($n=9$)	2026-08-24	F, ladder	$E>A$ frac	$p = 0.023$ combined
2^3 completion	2026-08-24	F, ext2 partial	$M\times R > 0$	not supported ($p = 0.47$)
Frontier scoping	2026-08-26	all above	estimation only	operators $\approx$ null
Family hint	2026-08-27	all above	optimize-once-given	yes as code, no as words
27B proposer (v)	2026-09-01	all above	wall holds vs breaks	holds (0.3876)

Table 3: Order of registrations and looks. The nine-problem primary analysis is a two-look sequence (discovery + prospective extension), not an independent confirmation.

9 Limitations

Nine problems is still a small universe of construction families; the full 2^3 isolates interactions, but at $n=9$ no per-term effect survives correction, so the operator decomposition (memory largest) is directional, not statistically resolved, and the collapse asymmetry is descriptive; budgets are equalized in proposals and verifier evaluations (120 of each per run; duplicate rejection in the repulsion arms happens after verification, since the signature requires the construction, and discards 20–30 verified candidates per run from the islands); the operators’ extra calls are measured at one per island-generation per operator (~ 220 characters per call, an overhead of 1–3% of proposer output characters; input-token accounting is incomplete, so budgets are matched in proposals and verifications, not demonstrably in tokens), and the arms differ in candidates surviving dedup into the islands (per run: A 108, B 95, C 93, D 72, BC 94, BD 65, CD 73, E 63), so sample efficiency here means verifier-evaluation efficiency; the diversity metric is related to the repulsion operator’s own filter; the behaviour signature is a hash of the returned construction after rounding, with no canonicalization for symmetries (permutation, rotation, reflection), so div can count symmetric variants as distinct, its rounding sensitivity is untested, and the D filter and the div metric share this signature (a retrospective AST-structure metric, invariant to renaming and independent of that signature, is directionally consistent on the runs with logged code: D-on arms 0.67–0.87 distinct structures per valid candidate against 0.38–0.39 for A and B on the matched problems; paired D–A +0.27 at the $n=3$ resolution floor); the nine problems (six chosen before any runs, three added by the pre-registered extension under the same criteria) satisfy criteria fixed before any runs (cheap exact verifier, short-function candidates, known headroom; the universe of 68 candidate pages (67 problems) and per-problem ratings are in the released reconnaissance table) but were not sampled randomly, so the exact tests quantify sign consistency within this selected suite and do not license random-sample generalization to the repository or to construction problems generally; because the problems and their literature are public, proposer performance does not distinguish retrieval of pretrained construction knowledge from de novo discovery (the generation-zero minimax program is the extreme case); *frac* is floored at the seed by construction, signs are harmonized for minimization, and reference values are frozen at the repository commit recorded in the release; the scaling ladder is one domain; the API proposer is not bitwise reproducible and its reasoning effort was fixed low; comparisons against repository values compare a 2026 proposer with values obtained by 2025-era systems, so they locate our wall and never rank systems: no system-level head-to-head with ShinkaEvolve or AlphaEvolve is attempted, and the operators should be read as structuring search, not as a substitute for construction knowledge the proposer lacks.

10 Conclusion

Inside a fixed verified-search loop at office scale, the operator packages our generation studies isolated (a schematic notebook, a named obstacle, behavioural duplicate control) compose into the arm with the highest mean, E, at one collapse in eighteen runs; memory+repulsion was the only zero-collapse configuration, within 0.025 of E’s mean, and is the parsimonious recipe these data support. Repulsion alone enforces behavioural non-duplication by construction and was the best single package exactly where the valid manifold is wide; and on the one problem where we compared them, a frontier proposer reached a higher value with roughly $20\times$ fewer verifier evaluations. On beat-the-average all three proposers stalled below a multiscale reference whose value is a continuum limit; the note added in version 3 reads that plateau as the resolution limit of the grid, with the reference’s family already in the loop’s finals, and the other plateaus as candidate family boundaries not yet established. Whether crossing such a boundary requires what we have informally called an idea was put to a registered test on the flagship problem: the idea, handed in words, was followed and lost; handed as code, it was optimized, but our

best tested finite-grid instantiation remained below the plateau the models reach unaided. In the registered test the loop moved the idea it was handed, and its three no-hint runs produced no candidate matching the idea’s signature; our finite encoding of the idea was not sufficient to close the gap here. Whether the last mile of these searches is an idea, the question this program has carried informally, remains open, now with its first measured answer.

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A Reproducibility and data availability

Code: <https://github.com/RobertoOno/interrupting-the-loop>, the repository of the program, with the search harness (`scripts/frontier_search.py`), the problem verifiers (`src/creative_machine/domains/`), the batteries (`run_f*.sh`, `run_hint.sh`), the analyses (`frontier_analysis.py`, `analysis_23.py`), the generated appendices (`docs/APPENDIX_F*.md`, `APPENDIX_HINT.md`) and the dated laboratory notebook (`docs/PLANO.md`, in Portuguese) with the pre-registrations and amendments of the batteries through the family-hint test. Run data and the later notebook entries are in three Zenodo deposits. (i) The release `run-data-v1` of the repository, deposited with `run-data-v2` as [10.5281/zenodo.22714138](https://zenodo.org/record/22714138): every logged candidate (`history.jsonl`, `best.py`) of the factorial with its replicate and extension batteries, of the frontier scoping runs and of the family-hint test. (ii) The evidence deposit of the later programs of this series (programs 5–7), [10.5281/zenodo.22713807](https://zenodo.org/record/22713807): the ladder runs, rungs (i)–(iv) in `runs/frontier/record` and the proposer-generation rung (v) in `runs/frontier/record27` (with `run_record27.sh`); the scoping runs again (`runs/frontier/FP`); and the analysis behind the note in Section 5 (`docs/APPENDIX_P2_FAMILY.md`, `family_beatavg.py`, `bf_ladder_limit.py`, `bf_mixture_grid.py`) with its dated notebook entry. (iii) The evidence deposit of this paper, [10.5281/zenodo.23163299](https://zenodo.org/record/23163299): the frontier proposer’s later composed-arm run at $L=5000$ (`runs/frontier/claude/beatavg_E300`) and the notebook entries made after version 1 that the other deposits lack (the pre-registration, amendments and single read of rung (v), and the entries behind the equal-grid comparison). Together the three deposits hold every candidate of every reported run. Proposing used `Qwen3-Coder-30B-A3B-Instruct` (8-bit, MLX) locally, `Qwen3.8-27B` served with vLLM on a rented H100 for rung (v), and `anthropic.claude-opus-5` on Amazon Bedrock for the frontier comparison.

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